Exam Date & Time: 28-Dec-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

END SEMESTER MAKE-UP EXAMINATION - DEC/JAN 2025

Pattern Recognition [BME 4068]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

Answer all questions.
Assume any missing data with proper reasoning.

1A) What is linear discriminant analysis? Explain all the steps in linear discriminant analysis. (4)

1B) As per the dataset shown below, apply a linear regression algorithm to predict y_i given x_i (3)

X_i (Week)	Y_i (Sales in thousands)
1	1.2
2	1.8
3	2.6
4	3.2
5	3.8

1C) What is ensemble learning? Illustrate the different advantages and specific real-life applications where ensemble learning methods can be used. (3)

2A) Given a one-dimensional dataset {1, 5, 8, 10, 2}, apply agglomerative clustering with complete link (taking Euclidean distance as the distance metric) to establish hierarchical group relationships. By using a cutting threshold of 5, how many clusters are there? (4)

2B) Illustrate the different steps involved in training a typical backpropagation in a neural network algorithm. (4)

2C) You roll six six-sided dice. Determine the probability that exactly two dice show the number 4, exactly three dice show the number 5, and the remaining die shows the number 6? (2)

3A) Apply DBSCAN with a similarity threshold of 0.8 (using the similarity matrix) to the given data points with minPts≥2 (minimum required points in a cluster). Also, highlight all the core, border, and noise (outliers) in the table as per the similarity matrix given below. (4)

	P1	P2	P3	P4	P5
P1	1	0.1	0.41	0.55	0.35
P2	0.1	1	0.64	0.47	0.98
P3	0.41	0.64	1	0.44	0.85
P4	0.55	0.47	0.44	1	0.76
P5	0.35	0.98	0.85	0.76	1

3B) Implement AND function using perceptron networks for bipolar inputs and targets. Set the initial weights and the threshold to 0 and the learning rate to 1. The truth table for AND function with bipolar inputs and targets is given below: (4)

X_1	X_2	t
1	1	1
1	-1	-1
-1	1	-1
-1	-1	-1

3C) What is k-fold cross-validation? Illustrate the advantages of k-fold cross-validation over traditional train-test split. (2)

4A) Given the dataset with two features, apply Principal Component Analysis (PCA) to reduce the dimensionality from two to one. (4)

Feature	Ex. 1	Ex. 2	Ex. 3	Ex. 4
X_1	4	8	13	7
X_2	11	4	5	14

4B) What is the bias-variance trade-off? Explain with an example. (3)

4C) Given the following training instances, each having two attributes (X_1 and X_2), compute the class labels for test instances t1 = (3, 7) using the 3-nearest neighbors technique (k = 3) (3)

Training Instance	X_1	X_2	Output
I_1	7	7	0
I_2	7	4	0
I_3	3	4	1
I_4	1	4	1

5A) Using the Support Vector Machine (SVM) algorithm, compute the hyperplane with a maximum margin for the following data: (5)

X_1	X_2	Class
2	2	-1
4	5	+1
7	4	+1

5B) Explain the different steps in training a Hopfield net algorithm. (3)

5C) Explain the different steps in the gradient descent algorithm with a real-life example. (2)

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